Week 09 teacher guide

Data, center, and spread

Keep this guide beside PowerPoint slides 299-334. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 057. Saturday | pages 223-227 | slides 299-317
Data displays and center

Day 058. Sunday | pages 228-240 | slides 318-334
Spread and box plots

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Data, center, and spread

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Weekly check

This week’s idea

Data vary. Center describes a typical value, and spread describes the variation. A display makes both easier to see.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Data displays and center

Day 057, pages 223-227 | PowerPoint slides 299-317

Goal: Read data displays and use mean or median to describe a typical value.

Set out: small counters, grid paper, ruler.

Check first: Ordering numbers, division, and reading a scale.

Pacing: Retrieval 5; launch 5; explore 15; discuss 10; synthesis 5; exit 5; preview 5; slack 10. Total 60 min.

Explore: what to watch for

Uses one dot per observation, including repeats. Why it matters: Preserves data frequency. Tutor move: Check that dot count equals data count.

Orders values before finding the middle. Why it matters: Makes median a position measure. Tutor move: Ask which values are on each side.

Uses redistribution or sum divided by count for mean. Why it matters: Both describe fair sharing. Tutor move: Accept both and relate the moved counters to the calculation.

Likely misconceptions

Omits repeated values in a dot plot. Why: Treats the display as a list of categories. Tutor move: Cross out each raw value after plotting one dot.

Chooses one middle value in an even set. Why: Uses an odd-set rule. Tutor move: Mark both middle values and find their midpoint.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 299. Week 9, Saturday

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Read data displays and use mean or median to describe a typical value.

Ask: What do you already know about this idea?

Expect: Ordering numbers, division, and reading a scale.

If stuck: Check that dot count equals data count.

Slide 300. Today

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A statistical question expects different answers. Mean shares the total equally; median locates the ordered middle.

If stuck: Read the learning goal together.

Slide 301. Recall | 3 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: (1, 2), (3, 6), (5, 10); 2: 88 square cm; 3: 5 units; 4: x = 7

If stuck: Give time to draw before giving a hint.

Slide 302. Recall: answers | 2 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: (1, 2), (3, 6), (5, 10); 88 square cm; 5 units; x = 7

If stuck: Rebuild one missed item with its model.

Slide 303. Launch | 5 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: How does your fair-share model match total divided by count?

Expect: P1: The shelf question; page counts can vary across books.

If stuck: Check that dot count equals data count.

Slide 304. Explore | 3 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: How does your fair-share model match total divided by count?

Expect: P2: Seven dots; counts 1, 2, 1, 2, 1 at 1 through 5. Symmetric around 3.

If stuck: Check that dot count equals data count.

Slide 305. The model | 2 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: The seven values balance around 3.

If stuck: Check that dot count equals data count.

Slide 306. Required practice | 2 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: How does your fair-share model match total divided by count?

Expect: P3: Bin counts: 3, 3, 2

If stuck: Check that dot count equals data count.

Slide 307. Required practice | 2 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: How does your fair-share model match total divided by count?

Expect: P4: Mean 5; deviations −3, −1, 0, 4 sum to 0; P5: (a) 4; (b) 5

If stuck: Check that dot count equals data count.

Slide 308. Required practice | 2 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: How does your fair-share model match total divided by count?

Expect: P6: Mean: 4 to 8. Median: 4 to 4. The median changes less.; P7: 9

If stuck: Check that dot count equals data count.

Slide 309. Watch the thinking | 2 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 5

If stuck: Return to the matching unit in the model.

Slide 310. Your turn | 2 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 6; 4.5

If stuck: Compare with the previous example without copying its numbers.

Slide 311. Compare methods | 5 min

Topic: Data displays and center

Student: Day 057, pages 223-227

Say: Partner compare: show the same problem in two ways.

Ask: Partner compare: How does your fair-share model match total divided by count? Agree or disagree: A very large value always moves the median as much as the mean. Use our data.

Expect: Accept drawings and equations that preserve the same quantities and units. Orders values before finding the middle.

If stuck: Ask which values are on each side.

Slide 312. A claim to check | 5 min

Topic: Data displays and center

Student: Day 057, pages 223-227

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: Order 2, 3, 6, 9, then average 3 and 6. Median = 4.5.

If stuck: Put the values on separate cards and order them.

Slide 313. Words and meaning | 3 min

Topic: Data displays and center

Student: Day 057, pages 223-227

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: How many books did each classmate read? The mean of 2 and 6 is 4. The median of 1, 4, 8 is 4. A bar can count heights from 120 to under 130 cm.

If stuck: Read one definition and draw its example.

Slide 314. In your own words | 2 min

Topic: Data displays and center

Student: Day 057, pages 223-227

Say: Use your own words and one example.

Ask: Partner compare: How does your fair-share model match total divided by count?

Expect: A statistical question expects different answers. Mean shares the total equally; median locates the ordered middle.

If stuck: Point to the model and name what each part represents.

Slide 315. Exit ticket | 5 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: Ordered: 2, 4, 5, 6, 8, 11. Median 5.5; mean 6.

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 316. The week ahead | 5 min

Topic: Data displays and center
Student: Day 057, pages 223-227

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 317. Tutor reference

Topic: Data displays and center
Student: Day 057, pages 223-227

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: Ordered: 2, 4, 5, 6, 8, 11. Median 5.5; mean 6.

Got it: Orders all six values and gives median 5.5 and mean 6 with correct methods.

Almost there: Orders correctly and uses both middle values, but makes one arithmetic slip.

Needs re-teaching: Uses the unsorted middle, chooses one middle value, or divides the total by the wrong count.

Sunday. Spread and box plots

Day 058, pages 228-240 | PowerPoint slides 318-334

Goal: Compare how data vary using range, IQR, and mean absolute deviation.

Set out: data cards, ruler, number line.

Check first: Ordering data, mean, median, and absolute value.

Pacing: Lesson: retrieval 3, launch 3, explore 12, discuss 10, synthesis 3, exit 5, preview 4. Total 40 min. Separate same-day cumulative check: 30 min.

Explore: what to watch for

Splits the ordered list by count. Why it matters: Finds quartiles from data positions. Tutor move: Count the values in each half.

Uses distances for MAD. Why it matters: Prevents cancellation. Tutor move: Ask why every gap is nonnegative.

Explains spread with numbers or a plot. Why it matters: Both provide evidence beyond the center. Tutor move: Accept either and connect the displayed gaps to the measure.

Likely misconceptions

Splits the range into four equal lengths. Why: Confuses numerical intervals with data quarters. Tutor move: Place each value on a card and divide the cards.

Averages signed deviations. Why: Finds balance rather than distance. Tutor move: Replace each difference with its absolute value.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 318. Week 9, Sunday

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Compare how data vary using range, IQR, and mean absolute deviation.

Ask: What do you already know about this idea?

Expect: Ordering data, mean, median, and absolute value.

If stuck: Count the values in each half.

Slide 319. Today

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Center describes a typical value. Spread describes how far values are apart or away from the center.

If stuck: Read the learning goal together.

Slide 320. Launch | 3 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: Why can two sets share a mean but have different MADs?

Expect: P1: 9

If stuck: Count the values in each half.

Slide 321. Explore | 2 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: Why can two sets share a mean but have different MADs?

Expect: P2: Minimum 1; Q1 3; median 6; Q3 9; maximum 11

If stuck: Count the values in each half.

Slide 322. The model | 2 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: The median splits the ordered data. Quartiles split each half.

If stuck: Count the values in each half.

Slide 323. Required practice | 2 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: Why can two sets share a mean but have different MADs?

Expect: P3: 1, 3, 7, 11, 13; IQR 8

If stuck: Count the values in each half.

Slide 324. Required practice | 2 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: Why can two sets share a mean but have different MADs?

Expect: P4: Mean 5; distances 3, 1, 1, 3; MAD 2

If stuck: Count the values in each half.

Slide 325. Required practice | 2 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: Why can two sets share a mean but have different MADs?

Expect: P5: Upper quarter: 9 to 12. No; each quarter represents about 25% of the data.

If stuck: Count the values in each half.

Slide 326. Required practice | 2 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Partner compare: Why can two sets share a mean but have different MADs?

Expect: P6: Both means 5. MAD A = 1, MAD B = 3. B is more spread out.; P7: Range 10 to 30; IQR stays 4.5

If stuck: Count the values in each half.

Slide 327. Watch the thinking | 2 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 6

If stuck: Return to the matching unit in the model.

Slide 328. Your turn | 1 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 1; 1; 2

If stuck: Compare with the previous example without copying its numbers.

Slide 329. A claim to check | 10 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: The distances 3, 1, 1, 3. Their mean is 2.

If stuck: Measure each gap from the mean with a ruler.

Slide 330. In your own words | 3 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Use your own words and one example.

Ask: Partner compare: Why can two sets share a mean but have different MADs?

Expect: Center describes a typical value. Spread describes how far values are apart or away from the center.

If stuck: Point to the model and name what each part represents.

Slide 331. Exit ticket | 5 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: Q1 3; median 6; Q3 10; IQR 7

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 332. The week ahead | 4 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 333. Grade 6 cumulative check | 30 min

Topic: Spread and box plots
Student: Day 058, pages 228-240

Say: Use the printed check and keep the key out of view.

Ask: Read each question before starting.

Expect: Score by skill using the separate answer key.

If stuck: Let the student skip and return. Record unattempted items separately.

Slide 334. Tutor reference

Topic: Spread and box plots

Student: Day 058, pages 228-240

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: Q1 3; median 6; Q3 10; IQR 7

Got it: Gives Q1 3, median 6, Q3 10, IQR 7 and splits by data count.

Almost there: Splits the data correctly but makes one median or subtraction error.

Needs re-teaching: Splits the range into equal lengths, uses unordered halves, or cannot identify the middle two values.

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 059. Monday | pages 241-242

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Data and statistics: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 060. Tuesday | pages 243-244

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Data and statistics: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 061. Wednesday | pages 245-246

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 062. Thursday | pages 247-248

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Number lines and coordinates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 063. Friday | pages 249-250

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Data displays and center

Day 057, student workbook pages 223-227

R1 [E-RELATE] | student p. 223

Answer: (1, 2), (3, 6), (5, 10)

Reasoning: Multiply each input by 2.

R2 [G-SA] | student p. 223

Answer: 88 square cm

Reasoning: Two faces of each type: 2 × (8 + 12 + 24).

Watch for 48: Uses volume.

R3 [N-DIST] | student p. 223

Answer: 5 units

Reasoning: One coordinate stays fixed. Count the gap in the other coordinate.

R4 [E-ADD] | student p. 223

Answer: x = 7

Reasoning: Subtract 11 from both sides.

S [EXPLAIN] | student p. 223

Answer: Answers vary. A statistical question expects different answers. Mean shares the total equally; median locates the ordered middle.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [S-QUESTION] | student p. 224

Answer: The shelf question; page counts can vary across books.

Reasoning: The first asks for one fixed book. The second expects several different values.

Watch for Both, because both have numbers: Confuses numerical answers with variability.

P2 [S-DISPLAY] | student p. 224

Answer: Seven dots; counts 1, 2, 1, 2, 1 at 1 through 5. Symmetric around 3.

Reasoning: Stack one dot for each student at that student's number of books.

Watch for Five dots: Counts distinct values instead of observations.

P3 [S-DISPLAY] | student p. 224

Answer: Bin counts: 3, 3, 2

Reasoning: A boundary value such as 2 goes into the bin that starts at 2.

Watch for Counts 2 in both first and second bins: Overlaps the interval endpoints.

P4 [S-MEAN] | student p. 225

Answer: Mean 5; deviations −3, −1, 0, 4 sum to 0

Reasoning: There are 20 counters for four jars. Move four from the jar of 9 to fill the two low jars.

Watch for Mean 9: Chooses the largest value.

P5 [S-MEDIAN] | student p. 225

Answer: (a) 4; (b) 5

Reasoning: (a) 2, 4, 4, 7, 8. (b) 2, 3, 5, 5, 6, 9; average the two middle 5s.

Watch for (a) 4; (b) 6: Uses the displayed middle without ordering.

P6 [S-CENTER] | student p. 225

Answer: Mean: 4 to 8. Median: 4 to 4. The median changes less.

Reasoning: The outlier increases the total by 20, so the five-value mean increases by 4. The middle value stays 4.

Watch for Both centers increase by 20: Does not distinguish middle position from fair-share total.

P7 [S-MEAN] | student p. 226

Answer: 9

Reasoning: The total must be 24. The known scores total 15, leaving 9.

Watch for 6: Assumes every score must equal the mean.

F [S-MEDIAN] | student p. 226

Answer: 6; 4.5

Reasoning: Use the worked example in the slides to check each step.

D [S-MEDIAN] | student p. 226

Answer: Order 2, 3, 6, 9, then average 3 and 6. Median = 4.5.

Reasoning: Finds the displayed middle before ordering. Put the values on separate cards and order them.

E [S-MEDIAN] | student p. 227

Answer: Ordered: 2, 4, 5, 6, 8, 11. Median 5.5; mean 6.

Reasoning: The middle values are 5 and 6. Their mean is 5.5. The total is 36, and 36 ÷ 6 = 6.

Watch for Median 8: Averages the displayed middle values 11 and 5 without ordering.

Spread and box plots

Day 058, student workbook pages 228-232

P1 [S-RANGE] | student p. 228

Answer: 9

Reasoning: $12 - 3 = 9$.

Watch for 12: Uses only the maximum.

P2 [S-IQR] | student p. 228

Answer: Minimum 1; Q1 3; median 6; Q3 9; maximum 11

Reasoning: The halves are 1, 3, 4 and 8, 9, 11. Their medians are 3 and 9.

Watch for Q1 = 3.5: Includes the overall median in a half under a different convention.

P3 [S-IQR] | student p. 228

Answer: 1, 3, 7, 11, 13; IQR 8

Reasoning: The halves contain four values each. Average each half's two middle values.

Watch for Q1 = 4; Q3 = 10: Splits the numerical range into quarters.

P4 [S-MAD] | student p. 229

Answer: Mean 5; distances 3, 1, 1, 3; MAD 2

Reasoning: Average the nonnegative distances: $8 \div 4 = 2$.

Watch for MAD 0: Averages signed differences, which cancel.

P5 [S-BOX] | student p. 229

Answer: Upper quarter: 9 to 12. No; each quarter represents about 25% of the data.

Reasoning: The plot groups data by ordered positions, not equal numerical lengths.

Watch for 5 to 9 has more data because it is longer: Confuses spread with frequency.

P6 [S-SPREAD] | student p. 229

Answer: Both means 5. MAD A = 1, MAD B = 3. B is more spread out.

Reasoning: B's values lie farther from the shared mean on average.

Watch for They have the same spread because their means match: Uses center alone to describe a distribution.

P7 [S-SPREAD] | student p. 230

Answer: Range 10 to 30; IQR stays 4.5

Reasoning: Q1 is 3.5 and Q3 is 8 in both sets. The maximum changes, but these middle-half markers do not.

Watch for IQR increases by 20: Assumes every spread measure reacts the same way.

F [S-MAD] | student p. 230

Answer: 1; 1; 2

Reasoning: Use the worked example in the slides to check each step.

D [S-MAD] | student p. 230

Answer: The distances 3, 1, 1, 3. Their mean is 2.

Reasoning: Signed deviations cancel instead of measuring distance. Measure each gap from the mean with a ruler.

S [EXPLAIN] | student p. 231

Answer: Answers vary. Center describes a typical value. Spread describes how far values are apart or away from the center.

Reasoning: Accept an accurate explanation supported by a matching example.

E [S-IQR] | student p. 232

Answer: Q1 3; median 6; Q3 10; IQR 7

Reasoning: Lower half: 1, 3, 3, 5. Upper half: 7, 9, 11, 13. Use each half's median.

Watch for Q1 4; Q3 10; IQR 6: Divides the min-to-max interval into four equal lengths.

Grade 6 cumulative check

Day 058, student workbook pages 233-236

Score by skill, including the requested reasoning. Use 80% or more as secure, 50-79% as developing, and below 50% as a review priority. Revisit a missed model, then check again with a different problem. This tutoring check does not determine school placement.

1 [R-RATIO] | student p. 233

Answer: 5 : 2; 2 : 5

Reasoning: Keep the requested first color first.

2 [R-TABLE] | student p. 233

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

3 [R-RATE] | student p. 233

Answer: 24 miles

Reasoning: Five minutes corresponds to two miles. Twelve five-minute groups make 24 miles.

4 [P-PART] | student p. 233

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

5 [P-WHOLE] | student p. 233

Answer: 40 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 3.60 shells: Taking 30% of the known part does not recover the whole.

6 [F-DIV] | student p. 233

Answer: $1\ \frac{1}{6}$ groups

Reasoning: A group is six eighths. Seven eighths is seven sixths of that group.

7 [F-DIV] | student p. 234

Answer: $\frac{5}{24}$ liter

Reasoning: Split each sixth into four equal parts. Each cup gets five twenty-fourths.

8 [F-DIV] | student p. 234

Answer: $\frac{2}{5}$

Reasoning: Two and one half is five halves. Five halves times two fifths equals one.

9 [D-MULT] | student p. 234

Answer: $7.35

Reasoning: 245 cents times three equals 735 cents.

10 [D-DIV] | student p. 234

Answer: 1.2

Reasoning: Ninety-six tenths shared among eight gives twelve tenths.

11 [N-DIV] | student p. 234

Answer: 42

Reasoning: Thirty-two times forty is 1,280. Two more groups add 64.

12 [E-DIST] | student p. 234

Answer: 2x + 14

Reasoning: Multiply x and 7 by 2.

Watch for 2x + 7: Multiplies only the first term.

13 [E-ADD] | student p. 235

Answer: x = 14

Reasoning: Subtract 9 from both sides.

14 [E-MULT] | student p. 235

Answer: x = 16

Reasoning: 3 equal parts total 12. Find one part, then all 4.

15 [G-PARA] | student p. 235

Answer: 36 square cm

Reasoning: Use base × perpendicular height: 9 × 4.

Watch for 45: Uses the slanted side.

16 [G-TRI] | student p. 235

Answer: 18 square cm

Reasoning: Half of 9 × 4 is 18.

Watch for 36: Omits the half.

17 [G-SA] | student p. 235

Answer: 60 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 48.

Watch for 54: Omits one triangular end.

18 [N-ORDER] | student p. 235

Answer: −7, −1, 2

Reasoning: Read the numbers from left to right on a number line.

19 [N-REFLECT] | student p. 236

Answer: (3, 4)

Reasoning: Keep the coordinate along the y-axis fixed; change the other sign.

20 [E-INEQ] | student p. 236

Answer: $x \geq 9$; yes

Reasoning: Use a closed circle if equality is allowed; otherwise use an open circle.

21 [S-MEAN] | student p. 236

Answer: 5

Reasoning: Total 20 divided by 4 values gives 5.

22 [S-MEDIAN] | student p. 236

Answer: 5

Reasoning: Order: 1, 3, 5, 7, 9. Use the middle value or mean of two middles.

23 [S-IQR] | student p. 236

Answer: Q1 3.0; Q3 9.0; IQR 6

Reasoning: Split the ordered list into equal-count halves. Subtract the two half-medians.

24 [S-MAD] | student p. 236

Answer: 2

Reasoning: Mean 4. Average the distances from 4.

Watch for 0: Averages signed differences instead of distances.

Study guide: Try it

Day 058, student workbook pages 237-240

1 [S-MEAN] | student p. 240

Answer: 5

Reasoning: Total 20 divided by 4 values gives 5.

2 [S-MEAN] | student p. 240

Answer: 4

Reasoning: Total 16 divided by 4 values gives 4.

3 [S-RANGE] | student p. 240

Answer: 9

Reasoning: 11 − 2 = 9.

4 [S-RANGE] | student p. 240

Answer: 12

Reasoning: 15 − 3 = 12.

5 [S-RANGE] | student p. 240

Answer: 10

Reasoning: 10 − 0 = 10.

Daily practice

Day 059, student workbook pages 241-242

1 [N-FACT] | student p. 241

Answer: 33

2 [N-FACT] | student p. 241

Answer: 77

3 [N-FACT] | student p. 241

Answer: 44

4 [N-FACT] | student p. 241

Answer: 99

5 [S-MEAN] | student p. 241

Answer: 4

Reasoning: Total 16 divided by 4 values gives 4.

6 [S-DISPLAY] | student p. 241

Answer: 3

Reasoning: Count every occurrence of 3.

7 [S-MEAN] | student p. 241

Answer: 6

Reasoning: Total 24 divided by 4 values gives 6.

8 [S-DISPLAY] | student p. 241

Answer: 3

Reasoning: Count every occurrence of 2.

9 [S-MEAN] | student p. 242 | Built-in review: Data and statistics

Answer: 5

Reasoning: Total 20 divided by 4 values gives 5.

10 [S-MEDIAN] | student p. 242 | Built-in review: Data and statistics

Answer: 5

Reasoning: Order: 1, 3, 5, 7, 9. Use the middle value or mean of two middles.

11 [S-MEDIAN] | student p. 242 | Built-in review: Data and statistics

Answer: 5

Reasoning: Order: 2, 4, 6, 8. Use the middle value or mean of two middles.

12 [S-MEAN] | student p. 242 | Built-in review: Data and statistics

Answer: 7

Reasoning: The total is 20; subtract the known total 13.

13 [S-MEDIAN] | student p. 242

Answer: Order 2, 3, 6, 9, then average 3 and 6. Median = 4.5.

Watch for 4: Finds the displayed middle before ordering.

Daily practice

Day 060, student workbook pages 243-244

1 [F-EQUIV] | student p. 243

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 243

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 243

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 243

Answer: 0.125; 12.5%

5 [S-MAD] | student p. 243

Answer: 3

Reasoning: Mean 6. Average the distances from 6.

Watch for 0: Averages signed differences instead of distances.

6 [S-SPREAD] | student p. 243

Answer: Both means 6. Range A 2; range B 6. B is more spread out.

Reasoning: Find the center, then compare the endpoint gaps.

7 [S-IQR] | student p. 243

Answer: Q1 3; Q3 10; IQR 7

Reasoning: Split the ordered list into equal-count halves. Subtract the two half-medians.

8 [N-ABS] | student p. 243

Answer: 12 units from zero

Reasoning: Count the unit steps between the number and zero.

9 [S-RANGE] | student p. 244 | Built-in review: Data and statistics

Answer: 9

Reasoning: 11 − 2 = 9.

10 [S-IQR] | student p. 244 | Built-in review: Data and statistics

Answer: Q1 3.0; Q3 9.0; IQR 6

Reasoning: Split the ordered list into equal-count halves. Subtract the two half-medians.

11 [S-MAD] | student p. 244 | Built-in review: Data and statistics

Answer: 2

Reasoning: Mean 4. Average the distances from 4.

Watch for 0: Averages signed differences instead of distances.

12 [S-SPREAD] | student p. 244 | Built-in review: Data and statistics

Answer: Both means 4. Range A 2; range B 6. B is more spread out.

Reasoning: Find the center, then compare the endpoint gaps.

13 [S-MAD] | student p. 244

Answer: The distances 3, 1, 1, 3. Their mean is 2.

Watch for 0: Signed deviations cancel instead of measuring distance.

Daily practice

Day 061, student workbook pages 245-246

1 [N-SCALE] | student p. 245

Answer: 112

2 [N-SCALE] | student p. 245

Answer: 11

3 [N-SCALE] | student p. 245

Answer: 23

4 [N-SCALE] | student p. 245

Answer: 33

5 [S-RANGE] | student p. 245

Answer: 10

Reasoning: $10 - 0 = 10$.

6 [S-MAD] | student p. 245

Answer: $\frac{3}{2}$

Reasoning: Mean 5. Average the distances from 5.

Watch for 0: Averages signed differences instead of distances.

7 [N-ABS] | student p. 245

Answer: 3 units from zero

Reasoning: Count the unit steps between the number and zero.

8 [G-SA] | student p. 245

Answer: 84 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 72.

Watch for 78: Omits one triangular end.

9 [R-TABLE] | student p. 246

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

10 [F-DIV] | student p. 246

Answer: 6 pieces

Reasoning: Each meter contains 3 equal pieces. Multiply by 2 meters.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

11 [N-FACT] | student p. 246

Answer: 28

12 [G-VOL] | student p. 246

Answer: 84 cm^3

13 [S-MEAN] | student p. 246 | Built-in review: Weekly puzzle

Answer: 2, 3, 5, 8, 12 and 2, 4, 5, 7, 12

Reasoning: The total must be 30. Known values 2, 5, 12 total 19. The remaining values must total 11, with one below 5 and one above.

Watch for 2, 5, 5, 6, 12: Repeats a number despite the distinct-value condition.

Daily practice

Day 062, student workbook pages 247-248

1 [N-EST] | student p. 247

Answer: 1100

2 [N-EST] | student p. 247

Answer: 12

3 [N-EST] | student p. 247

Answer: About 100

4 [N-EST] | student p. 247

Answer: About 20

5 [S-RANGE] | student p. 247

Answer: 10

Reasoning: $10 - 0 = 10$.

6 [S-MAD] | student p. 247

Answer: $\frac{3}{2}$

Reasoning: Mean 5. Average the distances from 5.

Watch for 0: Averages signed differences instead of distances.

7 [S-DISPLAY] | student p. 247

Answer: 2

Reasoning: Count every occurrence of 4.

8 [G-SA] | student p. 247

Answer: 60 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 48.

Watch for 54: Omits one triangular end.

9 [N-ORDER] | student p. 248 | Built-in review: Number lines and coordinates

Answer: −6, −3, 1

Reasoning: Read the numbers from left to right on a number line.

10 [N-ABS] | student p. 248 | Built-in review: Number lines and coordinates

Answer: 5 units from zero

Reasoning: Count the unit steps between the number and zero.

11 [N-REFLECT] | student p. 248 | Built-in review: Number lines and coordinates

Answer: (2, 5)

Reasoning: Keep the coordinate along the x-axis fixed; change the other sign.

12 [N-DIST] | student p. 248 | Built-in review: Number lines and coordinates

Answer: 5 units

Reasoning: One coordinate stays fixed. Count the gap in the other coordinate.

13 [S-IQR] | student p. 248

Answer: Q1 2.5; Q3 9.5; IQR 7

Reasoning: Split the ordered list into equal-count halves. Subtract the two half-medians.

14 [S-MAD] | student p. 248

Answer: The distances 3, 1, 1, 3. Their mean is 2.

Watch for 0: Signed deviations cancel instead of measuring distance.

15 [R-PERCENT] | student p. 248

Answer: $30

16 [E-EQ] | student p. 248

Answer: $x = 4$

Reasoning: Divide both sides by 3, then subtract 2.

Weekly check

Day 063, student workbook pages 249-250

1 [S-RANGE] | student p. 249

Answer: 10

Reasoning: $10 - 0 = 10$.

2 [S-MAD] | student p. 249

Answer: $\frac{3}{2}$

Reasoning: Mean 5. Average the distances from 5.

Watch for 0: Averages signed differences instead of distances.

3 [S-DISPLAY] | student p. 249

Answer: 2

Reasoning: Count every occurrence of 4.

4 [S-MEAN] | student p. 249

Answer: 10

Reasoning: The total is 28; subtract the known total 18.

5 [G-TRI] | student p. 249

Answer: 18 square cm

Reasoning: Half of 9×4 is 18.

Watch for 36: Omits the half.

6 [N-DIST] | student p. 250

Answer: 7 units

Reasoning: One coordinate stays fixed. Count the gap in the other coordinate.

7 [E-INEQ] | student p. 250

Answer: $x \leq 12$; yes

Reasoning: Use a closed circle if equality is allowed; otherwise use an open circle.

8 [R-TABLE] | student p. 250

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

9 [F-DIV] | student p. 250

Answer: 6 pieces

Reasoning: Each meter contains 3 equal pieces. Multiply by 2 meters.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

10 [S-MEDIAN] | student p. 250

Answer: Order 2, 3, 6, 9, then average 3 and 6. Median = 4.5.

Watch for 4: Finds the displayed middle before ordering.